

Robust rPPG Monitoring and Motion Artifact Cancellation via Dual-Modality: RGB Video and Wearable PPG Validation

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Project Goal(s):

The primary goal is to develop a non-invasive, "nearable" system for Remote Photoplethysmography (rPPG) that accurately estimates heart rate from standard RGB video, specifically addressing the challenge of motion interference. Our objectives include:

- Developing a pipeline to extract heart rate signals by analyzing subtle skin color variations.
- Implementing Head Pose Estimation to ensure data is only processed when the subject is in an ideal position.
- Designing a Frequency Cancellation algorithm to filter out motion artifacts caused by rhythmic head movements (e.g., nodding while listening to music) while maintaining the integrity of the rPPG signal.
- Validating the system's accuracy by comparing "nearable" camera data against a Wearable PPG as the gold standard ground truth.

Health/Physiological Data to Measure and Analyze

- Heart Rate (BPM): Extracted via rPPG from the green color channel of the RGB sensor.
- Head Pose Coordinates: (Yaw, Pitch, Roll) to assess measurement reliability.
- Motion Magnitude: Calculated using Dense Optical Flow to quantify and cancel physical movement noise.
- Ground Truth Data: Heart rate obtained from a wearable PPG.

Sensors Required

- **Standard RGB Camera:** Used for face detection, skin segmentation, and rPPG extraction.
- **CMS50D Pulse Oximeter:** Used for HR, SpO2, pulse wave from the subject's finger